KK-L-156D
August 26, 1976
SUPERSEDING
Fed. Spec. KK-L-156C
April 11, 1968

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, HARNESS, VEGETABLE TANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for vegetable tanned cattlehide harness leather (see 6.1).

1.2 Classification. The leather shall be furnished in the following types and classes, as specified (see 6.2):

Type I - Sides.
Type II - Backs.
Type III - Bends.
Type IV - Double shoulders.
Class 1 - Mildew-resistant treated.
Class 2 - Not mildew-resistant treated.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-311 - Leather, Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 8330

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Military Specification:

MIL-L-43432 - Leather, Packaging and Packing of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The leather shall be vegetable tanned, full grain or corrected grain cattlehide. Corrected grain leather shall be full grain leather that has been lightly snuffed on the grain side to remove superficial blemishes. The flesh side shall be smooth and free of loose flesh and may be shaved or buffed to provide greater uniformity of thickness. Use of bull hides is prohibited (see 6.3).

3.2 Physical requirements.

3.2.1 Trim (form and area). The leather shall be furnished in the form of sides, backs, bends or double shoulders as required for the type specified (see 6.2) and shall be trimmed to conform with the definition of sides, backs, bends or double shoulders as listed under "Definition of Terms" in FED-STD-311. The area in terms of square feet shall be legibly marked on the flesh side of the leather. Sides, backs and bends shall be marked in the butt area; double shoulders shall be marked in the corner formed by the belly cut and the straight edge cut.

3.2.2 Color. The color of the leather shall be as specified (see 6.2). A light application of dye and pigment shall be employed when necessary to provide a uniform color on the grain side of the leather.

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3.2.3 Finish. Unless otherwise specified (see 6.2), the leather shall have a light application of finish on the grain side.

3.2.4 Weight and thickness. The leather shall be of the weight and thickness specified (see 6.2), and as shown in table I.

TABLE I. Weight and thickness

Weight	Thickness (ounces) 1/
Extra light	8 to 9-3/4
Light	10 to 11-3/4
Medium	12 to 13-3/4
Heavy	14 to 15-3/4
Extra heavy	16 and above

1/ 1 ounce = 1/64 inch.

3.2.5 Breaking force. At least 80 percent of the specimens tested shall meet the minimum breaking force values specified in table II, when tested as specified in 4.3.

TABLE II. Breaking force

Thickness (ounces)	Breaking force minimum (pounds)
8 to 9-3/4	135
10 to 11-3/4	170
12 to 13-3/4	210
14 to 15-3/4	240
16 and above	260

3.2.6 Elongation. At least 80 percent of the specimens tested shall have elongation values no greater than 20 percent under the loads shown in table III when tested as specified in 4.3.

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TABLE III. Elongation load

Thickness (ounces)	Load (pounds)
8 to 9-3/4	110
10 to 11-3/4	140
12 to 13-3/4	170
14 to 15-3/4	200
16 and above	230

3.2.7 Cracking. The leather shall not crack when tested as specified in 4.3.

3.3 Chemical requirements.

3.3.1 The leather shall conform to the chemical requirements of table IV when tested as specified in 4.3.

TABLE IV. Chemical requirements

Characteristic	Minimum	Maximum
Chloroform-soluble material, percent 1/	15.0	35.0
Water-soluble material, percent 1/	--	18.0
Total ash, percent 1/	--	5.0
Lead, qualitative	2/	2/
Acidity (pH)	3.3	5.0

1/ Calculated on the moisture free basis.

2/ The leather shall be tested for lead only when pigments are used. The test results shall be negative.

3.3.2 Class 1. When class 1 is specified (see 6.2), the leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3 (see 6.4).

3.4 Workmanship. The finished leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with FED-STD-311 except when otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The defects found during the examination of the leather for visual and dimensional characteristics shall be in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one side, back, bend or double shoulder. The lot size shall be expressed in units or one side, back, bend or double shoulder. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more units of product. Separate examinations shall be performed for visual and dimensional examinations, but the same sample may be used for each examination.

4.2.2.1 Examination for visual characteristics.

Examine	Defect
Trim	Not as specified.
Color	Not as specified.

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Examine	Defect
Finish	Unless otherwise specified, grain side does not have application of finish.
Quality of leather	Grain not as specified. Raw streak. Flesh side not smooth or contains areas of loose flesh in excess of four square inches in total area. Stain, burn mark, cut, scratch, hole, brand, hip mark, rough shoulder, heavy vein, deep fat wrinkle, boney or flanky area, or other surface imperfections which impair the cutting value of more than 12 percent of the total area of the leather.

NOTE: Light, well healed, scratches or grub holes; light fat wrinkles; or slight stains shall not be classified as defects.

4.2.2.2 Examination for dimensional characteristics.

Examine	Defect
Area (expressed in terms of square feet)	Less than area marked on leather.
Marking	Omitted.
Thickness 1/	Not as specified.

1/ The thickness shall be determined in accordance with Method 1011 of FED-STD-311. No individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine whether packaging, packing and marking comply with the section 5 requirements of this specification.

4.3 End item testing. The methods of testing specified in FED-STD-311 wherever applicable and as listed in table V, shall be followed. Sampling procedures shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:

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a. More than 3 specimens fail to meet requirements for breaking force.
b. More than 3 specimens fail to meet requirements for elongation.
c. More than one specimen fails to meet requirements for cracking.
d. Any composite value fails to meet the specified requirement.

TABLE V. Tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.1	1/	-	-
Breaking force	3.2.5	2021	X	-
Elongation	3.2.6	2021	X	-
Cracking	3.2.7	4011 2/	X	-
Chloroform-soluble material	3.3.1	6341	-	X
Water-soluble material	3.3.1	6341	-	X
Total ash	3.3.1	6421	-	X
Lead, qualitative	3.3.1	6551	-	X
Acidity (pH)	3.3.1	6621	-	X
Paranitrophenol	3.3.2	6711	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The diameters of the mandrels shall be as follows:

Thickness of leather (ounces)	Diameter of mandrel (inches)
8 to 9-3/4	5/16
10 to 11-3/4	3/8
12 to 13-3/4	7/16
14 to 15-3/4	1/2
16 and above	9/16

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. The leather is primarily intended for use in safety belts and harness straps.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2 and 3.2.1).
(c) Color required (see 3.2.2).
(d) When finish is not required (see 3.2.3).
(e) Weight and thickness required (see 3.2.4).
(f) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(g) Marking required (see 5.3).

6.3 Bull hides. Bull hides are excluded from the cattlehide to be used for harness since they do not produce good-quality leather. A bull hide usually ranges in weight from 75 to more than 100 pounds in the green, salted state. It is characterized by very thick and rough neck and shoulders.

6.4 Paranitrophenol. Paranitrophenol has a tendency to leach out of the leather when the leather becomes wet and will stain items that are in intimate contact with treated leather. Consideration should be given to this factor when determining the class of leather to be specified.

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Air Force - 11

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User activity:

Army - EL

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

USDA-ARS
GSA-FSS
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